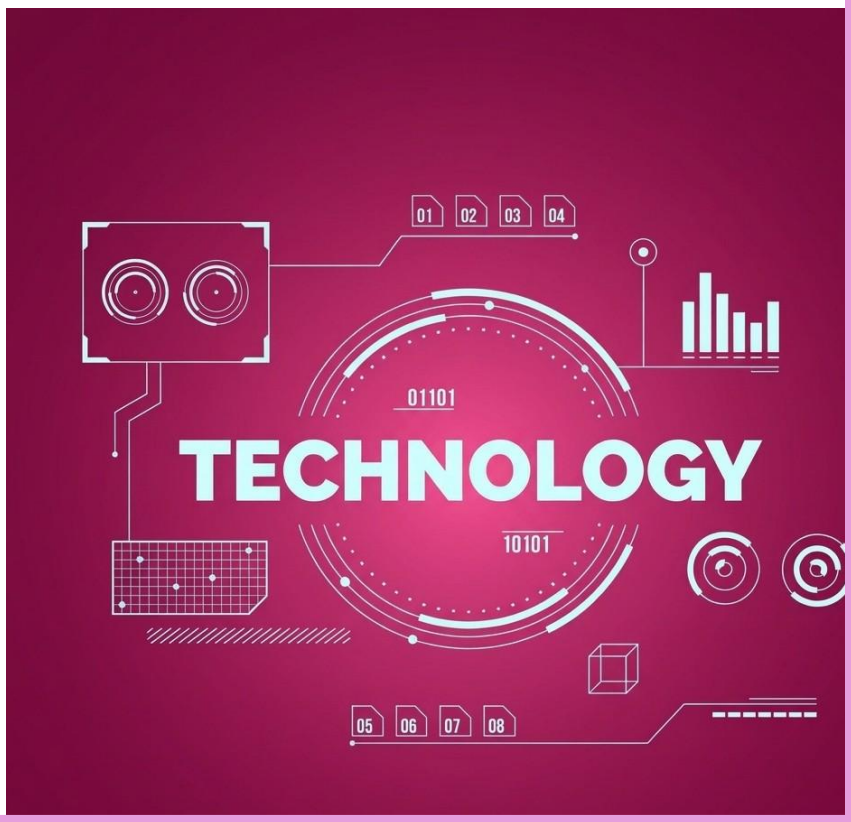


TASK 2

XITM6329 GROUP 3

DITM - GROUP 3

Tlotlang Ramatlotlo ST10445028

Praise Kapinga ST10452822

Zandile Nkomo ST10440938

Nomnotho Nshangase ST10462439

Kyle Bila ST10452906

Hardware requirements

Hardware requirements are the physical devices and equipment needed to build a working computer system or network. They describe the machines, cables, and devices that must be purchased, installed, and maintained to make the system function properly.

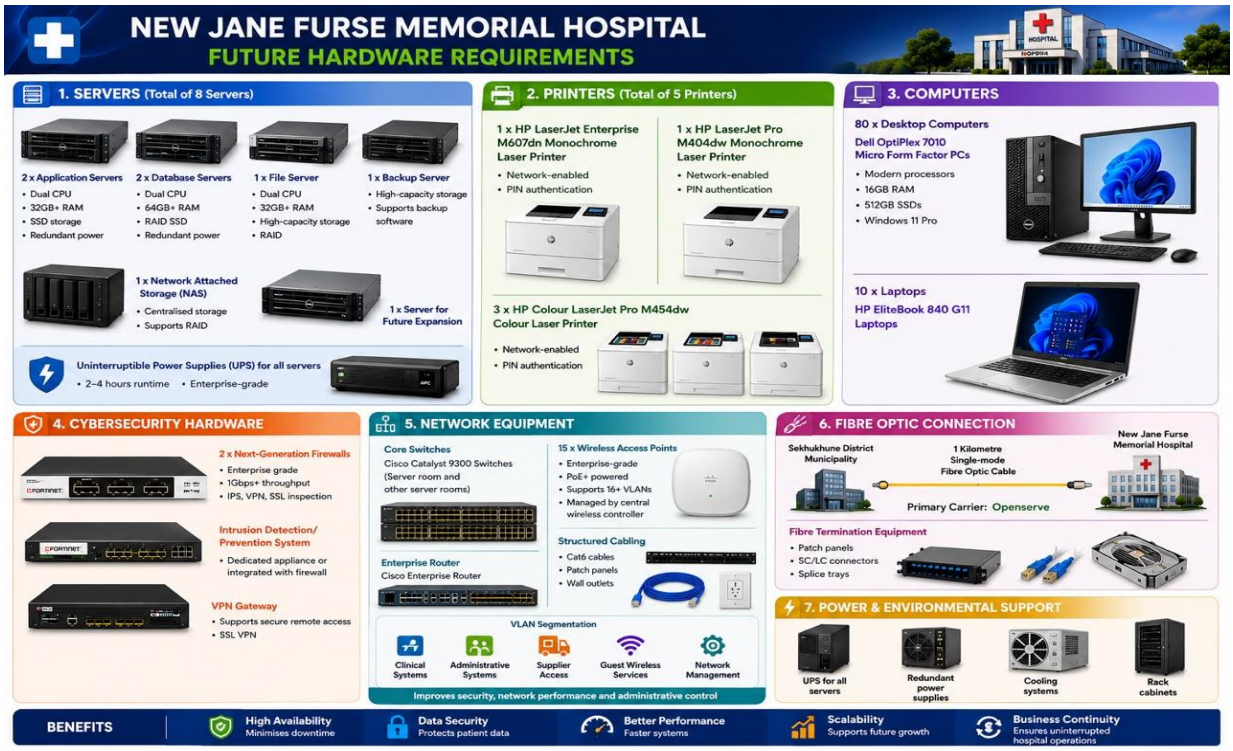


FIGURE 1: HARDWARE REQUIREMENTS FOR THE HOSPITAL

CURRENT	FUTURE
Servers not mentioned but there are challenges the hospital is facing with data storage and missing patient files.	For the future hardware components, we will have a total of 8 servers. We did this to make sure that if one server stopped working, the others could continue running the hospital management software without causing any interruptions. The type of server that will be used are 2 Application Servers with Dual CPU, 32GB and a RAM, SSD storage, redundant power, with 2 Database Servers Dual CPU, 64GB plus RAM, RAID SSD, redundant power, File Server with Dual CPU, 32GB plus RAM, high capacity storage, RAID, and Backup Server High capacity storage, supports backup software. Also have a Network Attached

	Storage Centralised storage, supports RAID and a Server for future expansion Uninterruptible Power Supplies (UPS) for all servers 2-4 hours runtime, enterprise-grade (Kurose, J.F and Ross, KW., 2021).
The hospital currently has only three printers available	We will install five printers in total to improve printing security and convenience. The printers that will be used are a HP LaserJet Enterprise M607dn Monochrome laser printer, network-enabled, PIN authentication also a HP LaserJet Pro M404dw Monochrome laser printer, network-enabled, PIN authentication and 3 x HP Colour LaserJet Pro M454dw Colour laser printer, network-enabled, PIN authentication (Kurose, J.F and Ross, KW., 2021).
Computers The hospital currently has only 15 very old computers	For the future we will install 80 desktop computers and 10 laptops to replace the outdated equipment and support all clinical and administrative staff. We will have 80 computers that are Dell OptiPlex 7010 Micro Form Factor PCs and have a modern processors 16GB RAM, 512GB SSDs, and Windows 11 Pro for improved performance and security.There will be 10 HP EliteBook 840 G11 laptops (Kurose, J.F and Ross, KW., 2021).
Cybersecurity Hardware The hospital currently has no security measures installed to secure the network against hackers.	The implemention of the cybersecurity controls to protect the hospital network and sensitive patient data. We deployed a next-generation firewall with intrusion detection and prevention capabilities, which monitors all incoming and outgoing network traffic and blocks suspicious activity. We will use a 2 x Next-Generation Firewalls Enterprise grade, 1Gbps+ throughput, IPS, VPN, SSL inspection also a Intrusion Detection/Prevention System Dedicated appliance or integrated with firewall and a VPN Gateway Supports secure remote access, SSL VPN (Kurose, J.F and Ross, KW., 2021).
The current network equipment at the hospital is not specified	For the future hardware components, we will install a high performance switches and routers to make the hospital network faster and more dependable. We used Cisco Catalyst 9300 switches in the server room and other server rooms around the hospital. For routing between the municipality and hospital, we installed a Cisco Enterprise

	Router that provides the dedicated fibre connection to the hospital. We also installed structured cabling throughout the hospital, including Cat6 cables, patch panels, and wall outlets. We installed a total of 15 wireless access points Enterprise-grade, PoE and a powered, supports 16 plus VLANs across the hospital, to eliminate the need for separate power outlets. These access points are managed by a central wireless controller. The network equipment was configured with VLANs to separate clinical systems, administrative systems, supplier access, guest wireless services, and network management traffic. This will improve security, network performance, and administrative control (Kurose, J.F and Ross, KW., 2021).
The hospital currently has no fibre optic connection to the municipality	For the future hardware components, will install a 1 kilometre of single-mode fibre optic cable from the Sekhukhune District Municipality to the New Jane Furse Memorial Hospital. Openserve will be used as the primary carrier for the fibre connection. We will install fibre termination equipment including patch panels, SC or an LC connectors, and splice trays to terminate and manage the fibre connections. We are going to use PVC or steel conduct along the entire 1 km route to protect the fibre cable from accidental cutting or vandalism. This fibre infrastructure provides a dedicated, high-speed, and reliable connection that significantly improves the hospital's connectivity and enables all future network services.
hospital does not have a dedicated, properly equipped server room	For the future hardware components, we are going to use established dedicated server rooms. We will install equipment racks in each server room to mount servers, switches, routers, and other network hardware. We will install appropriate cooling to prevent hardware failure due to overheating. We will also linstall environmental monitoring to track temperature and humidity levels. Physical access controls will be implemented to ensure only authorized ICT staff could access the server rooms. Uninterruptible

	power supplies be will installed to protect all critical equipment against power interruptions. This setup ensures the hardware operates reliably and remains protected from physical and environmental threats (Kurose, J.F and Ross, KW., 2021).

System and application software requirements

System software requirements

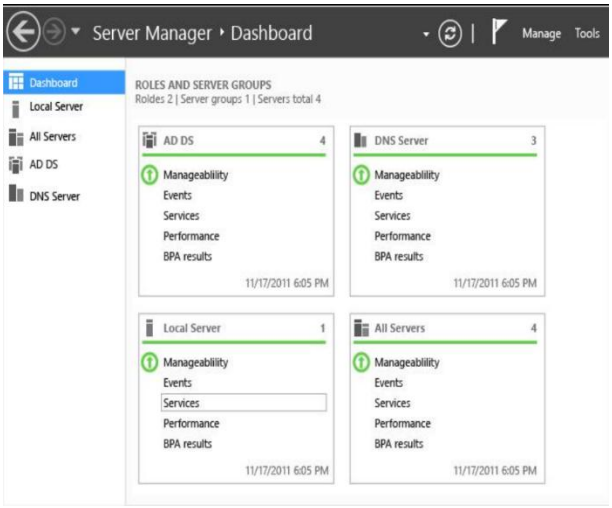


Figure 1: Windows Server Management System



Figure 2: Data Centre Servers

System software means the main software that manages computer hardware and provides a stable environment for all other applications to run. In the case of New Jane Furse Memorial Hospital, system software is really important because the hospital is moving from outdated, paper-based and unreliable systems to a modern digital network infrastructure. The system software will make sure that all computers, servers and network devices operate in an organised and safe way.

The primary system software required is a server operating system such as Windows Server. This is important because it allows centralised management of users, meaning that all user accounts, devices and permissions are controlled from one main system. This is done through Active

Directory. Active Directory is important in a hospital environment because it makes sure that only authorised staff such as doctors, nurses and administrative personnel can access sensitive patient information. This improves security and ensures proper control over hospital data.

Besides Windows Server, Linux Server operating systems may also be used for certain functions such as hosting databases, web applications and internal hospital systems. Linux is usually used in business environments because of its stability, high performance and lower cost. Using both Windows and Linux provides flexibility and makes the system more reliable.

Another important component of system software is backup and recovery software. This is important for the hospital because one of the major issues identified in the current system is missing patient files and poor data storage. Automated backup systems will make sure that data is copied daily and stored securely either on-site or off-site. In the event of system failure, cyberattacks, or accidental deletion, the hospital will be able to restore all important information quickly.

The hospital will also require virtualisation software such as VMware or Microsoft Hyper-V. Virtualisation allows multiple virtual servers to run on a single physical machine, reducing hardware costs and making the system more simple and effective. This is especially important for a hospital environment where multiple systems such as patient records, finance systems and supply chain systems must run at the same time.

Lastly, system monitoring software is needed to ensure that all servers and network devices are performing in the best way. This software will help ICT staff detect issues such as system overload, hardware failure or network downtime before they affect hospital operations. Since the hospital only has three ICT support staff, automated monitoring is important for maintaining the system's stability.

ICT Hardware, Software and Continuity Plan

New Jane Furse Memorial Hospital

This SOP governs the installation, configuration, operation, maintenance, security and recovery of the proposed hospital ICT environment. It is based on the stated requirements: eight servers, 80 desktop computers, 10 laptops, five secure network printers, dual next-generation firewalls, Cisco network equipment, 15 managed wireless access points and a dedicated 1 km Openserve fibre connection to Sekhukhune District Municipality.

The SOP supports confidentiality, integrity and availability of patient and hospital data. POPIA requires reasonable technical and organisational measures to prevent loss, unauthorised destruction, unlawful access and unlawful processing of personal information.

1. Document Control

Item	Detail
SOP title	ICT Hardware, System Software, Application Software and Business Continuity SOP
Organisation	New Jane Furse Memorial Hospital, Sekhukhune District
SOP owner	Hospital ICT Manager / Lead ICT Support Technician
Approval authority	Hospital CEO, District ICT Manager and Information Officer
Applies to	ICT staff, hospital staff, contractors, suppliers and approved service providers
Review date	Every six months, after an outage, cyber incident, major system change or annual continuity test
Related policies	Acceptable Use Policy, Access Control Policy, Network Security Policy, Backup Policy, Incident Response Policy, POPIA Privacy Policy and Supplier Access Policy

2. Purpose and Scope

The purpose of this SOP is to ensure that the hospital's technology resources are installed and used consistently, securely and sustainably. It also establishes a continuity plan for maintaining critical hospital services when there is a power failure, fibre outage, server failure, ransomware attack, device theft or application failure.

The SOP covers:

- Server and storage infrastructure.
- User computers, laptops and printers.
- Fibre, routers, switches, Wi-Fi and firewall equipment.
- System software, including Windows Server, Linux, Active Directory, virtualisation, backup and monitoring software.
- Application software, including Hospital Management System and Electronic Patient Record system.
- Data backups, user access, cybersecurity, maintenance and incident recovery.
- Secure external supplier access to Intralink services.

Hospital information systems should support interoperability requirements under South Africa's National Health Normative Standards Framework for Interoperability in Digital Health.

Part A: Hardware Requirements SOP

3. Hardware Standards

3.1 Server and storage environment

The hospital must use eight servers to support continuity and prevent one hardware failure from stopping critical systems.

Server type	Quantity	Minimum specification	Main function
Application server	2	Dual CPU, minimum 32 GB RAM, enterprise SSD storage, redundant power supplies	Hosts Hospital Management System, Electronic Patient Record and internal applications
Database server	2	Dual CPU, minimum 64 GB RAM, RAID SSD storage, redundant power supplies	Hosts patient, financial, supply-chain and application databases
File server	1	Dual CPU, minimum 32 GB RAM, high-capacity RAID storage, redundant power supplies	Secure storage for authorised documents, scanned patient files and departmental data
Backup server	1	Enterprise-grade server, high-capacity storage, backup software compatibility	Manages daily, weekly and monthly backups and recovery operations
Network Attached Storage	1	RAID-enabled central storage with encrypted shares	Stores backup copies and shared hospital files
Expansion server	1	Virtualisation-capable enterprise server	Supports future systems, expansion or disaster-recovery workloads

3.2 Server operating procedure

1. All servers must be installed in the dedicated server room and mounted in secure equipment racks.
2. Each server must have redundant power supplies connected to separate UPS circuits where available.
3. Servers must use RAID storage so that a disk failure does not immediately cause data loss.
4. Application and database servers must be configured in pairs where possible to provide failover capability.

5. The ICT team must install virtualisation software, such as VMware or Microsoft Hyper-V, to create separate virtual machines for different hospital services.
6. No server may be connected to the hospital network without firewall protection, antivirus/endpoint protection, secure configuration and approved administrator credentials.
7. Each server must be recorded in the ICT asset register, including make, model, serial number, IP address, operating system, installed applications, owner, warranty and backup status.
8. Only authorised ICT support staff may access server consoles, server-room racks or server administrator accounts.
9. Server logs must be retained and reviewed after security incidents or system failures.
10. The ICT team must test failover and restoration at least twice per year.

3.3 End-user devices

Hardware item	Quantity	Required specification	Primary users
Dell OptiPlex 7010 Micro PCs	80	Modern processor, 16 GB RAM, 512 GB SSD, Windows 11 Pro, Gigabit Ethernet	Clinical staff, administration, finance, supply chain and records staff
HP EliteBook 840 G11 laptops	10	Business-class laptop, encrypted storage, Wi-Fi, webcam and Windows 11 Pro	ICT staff, management and approved mobile clinical/administrative staff
Existing old computers	15	Must be assessed before reuse	May only be used if compatible with supported software and security controls

3.4 Device deployment procedure

1. ICT must assess each of the 15 existing computers before connecting it to the new network.
2. Devices that cannot support Windows 11 Pro, endpoint protection, encryption or current business applications must be retired.
3. Retired devices must be securely wiped before disposal, reuse or recycling.

4. Each new Dell desktop and HP laptop must receive an asset tag and be captured in the asset register.
5. Windows 11 Pro must be activated, patched and joined to the Active Directory domain before issue to a user.
6. Standard approved software must be installed through a controlled deployment process.
7. Local administrator rights must not be granted to ordinary users.
8. Laptops must have full-disk encryption, password protection, automatic screen locking and endpoint protection.
9. ICT must configure user devices for the correct network segment, such as clinical, administration or finance.
10. A department manager must sign the equipment issue form before a device is allocated.

3.5 Printer requirements and procedure

The hospital will operate five network-enabled printers using PIN or secure-print authentication.

Printer	Quantity	Location recommendation	Use
HP LaserJet Enterprise M607dn	1	Records, finance or central administration	High-volume secure monochrome printing
HP LaserJet Pro M404dw	1	Clinical administration or procurement office	General secure monochrome printing
HP Colour LaserJet Pro M454dw	3	Shared office areas	Colour reports, charts, training material and approved documentation

Printer SOP

- All printers must be network-connected through a controlled printer VLAN.
- Default printer passwords must be changed before deployment.
- PIN release or secure-print authentication must be enabled.

- Printers must not be placed in public areas where patient documents can be collected by unauthorised persons.
- Printed patient records must be collected immediately and placed in secure files.
- Printer firmware must be updated every quarter.
- Print logs should be reviewed when there is suspected data leakage or misuse.
- Scanning-to-email or scanning-to-network-folder functions must use approved, encrypted destinations only.

3.6 Network and cybersecurity hardware

Component	**Quantity**	**Requirement**	**Function**
Next-generation firewall	2	Enterprise grade, minimum 1 Gbps throughput, IPS, VPN, SSL inspection and high availability	Blocks unauthorised traffic and protects the hospital network
IDS/IPS	Integrated with firewall or dedicated appliance	Detects and blocks malicious activity	Cyberattack detection and prevention
VPN gateway	Integrated firewall or dedicated gateway	SSL VPN and multi-factor authentication	Secure access for ICT staff and authorised suppliers
Cisco Catalyst 9300 switches	As required	Managed, enterprise-grade, VLAN and PoE support	Core and access switching
Cisco enterprise router	1	Fibre routing, secure routing policies and traffic control	Connects hospital to municipality and Openserve carrier link

Component	Quantity	Requirement	Function
Wireless access points	15	Enterprise-grade, PoE+, central controller, supports 16+ VLANs	Provides managed hospital Wi-Fi
Wireless controller	1	Centrally manages access points, SSIDs and security policies	Wi-Fi administration
Fibre-optic cable	1 km	Single-mode fibre in PVC/steel conduit	Dedicated municipal-to-hospital connectivity
Fibre termination equipment	As required	Patch panels, SC/LC connectors and splice trays	Terminates and manages fibre link
UPS units	All critical equipment	Enterprise-grade, 2–4-hour runtime	Protects equipment during power failure

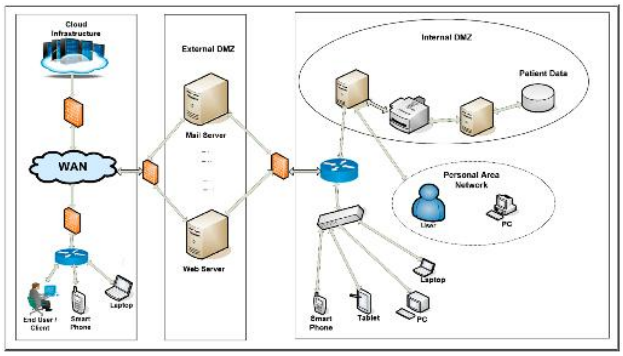


Figure1: Example of a Healthcare Cybersecurity Network Design. Firewalls, network segmentation, secure access controls and protected server zones reduce the risk of unauthorised access to sensitive healthcare systems and patient records.

3.7 Network segmentation procedure

ICT must configure VLANs to isolate users and systems. This limits the spread of malware and prevents unauthorised access.

VLAN	Authorised systems	Restrictions
Clinical VLAN	Electronic Patient Records, ward computers, pharmacy and laboratory systems	May access approved clinical systems only
Administration VLAN	HR, finance, management and office computers	Cannot access clinical databases directly unless authorised
Supply-chain VLAN	Procurement and inventory devices	Restricted to supply-chain applications
Supplier VPN VLAN	Approved remote suppliers	No direct access to internal hospital network or patient records
Guest Wi-Fi VLAN	Visitors and non-hospital devices	Internet access only; no access to internal systems
ICT Management VLAN	ICT support devices, monitoring tools and network administration	Restricted to authorised ICT administrators
Printer VLAN	Approved hospital printers	Controlled communication with print server only
Server VLAN	Application, database, backup and file servers	Strict firewall rules and limited access

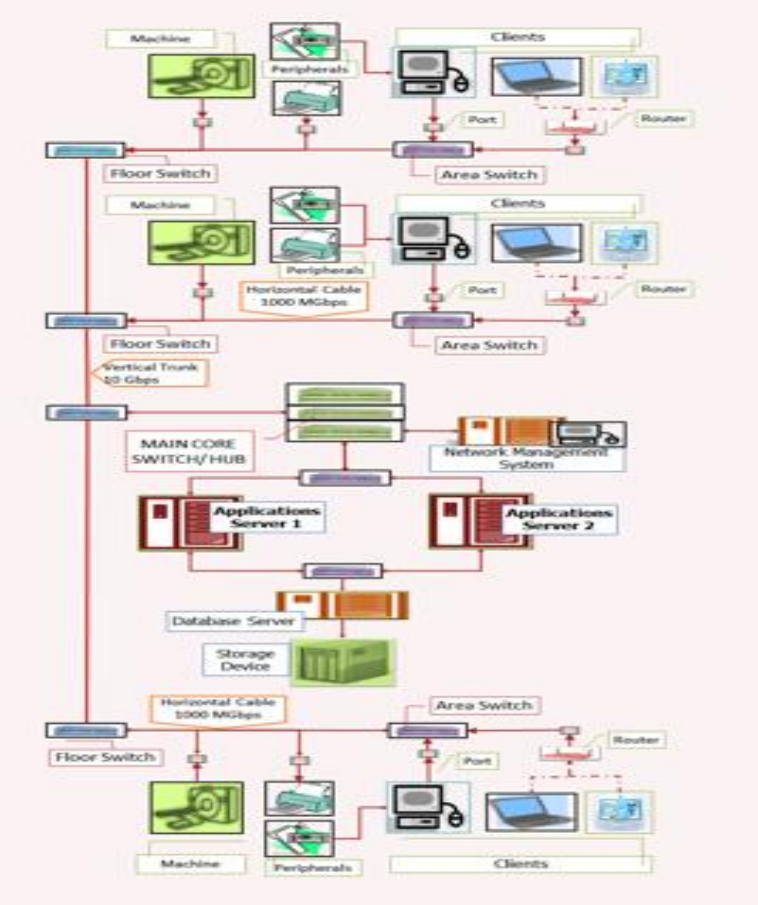


Figure 2: Example of a Hospital Network Topology. Hospital departments, workstations, printers, wireless access points, servers and storage are connected through a structured hierarchy of access switches and a core network. In the proposed solution, VLANs and firewall rules provide additional separation between clinical and non-clinical systems.

3.8 Server-room requirements

The hospital must establish dedicated server rooms that are physically secure and environmentally controlled.

The server room must include:

- Lockable equipment racks.
- Restricted physical access using keys, access cards or biometric access.
- CCTV monitoring where available.
- Cooling and ventilation to prevent overheating.
- Temperature and humidity monitoring.

- Smoke detection and appropriate fire-control equipment.
- UPS power protection for all critical equipment.
- Cable management, equipment labels and updated network diagrams.
- No food, drinks, paper storage or unauthorised equipment.
- A visitors' log for maintenance contractors and suppliers.



Figure 3: Example of a Secure Hospital Server Room with Server Racks and UPS Equipment. The planned New Jane Furse server room will contain locked racks, cooling, environmental monitoring, UPS protection, controlled physical access and enterprise servers.

Part B: System Software Requirements SOP

4. System Software Standards

System software creates the controlled environment in which computers, network devices and hospital applications operate.

System software	Requirement	Purpose
Windows Server	Supported version of Windows Server	Central server management, user management and application hosting
Active Directory	Installed and configured on redundant domain controllers	Creates accounts, groups, permissions and role-based access

System software	Requirement	Purpose
Linux Server	Supported enterprise Linux distribution where suitable	Database, web, monitoring or internal application hosting
VMware or Microsoft Hyper-V	Installed on virtualisation-capable servers	Runs multiple virtual servers efficiently
Backup and recovery software	Enterprise backup application with scheduling, encryption and restoration testing	Protects patient and operational records
Monitoring software	Server, network and storage monitoring with alerting	Helps three ICT staff detect failures early
Endpoint protection	Centralised antivirus/EDR solution	Protects desktops, laptops and servers against malware
Patch-management software	Central patching and reporting system	Keeps software and systems updated
Print-management software	Secure print, PIN release and print audit controls	Protects confidential printed information
Remote-support software	Approved, logged remote administration tool	Allows secure ICT support and troubleshooting

4.1 Windows Server and Active Directory SOP

1. Windows Server must be installed only on approved application, database, file and management servers.
2. At least two Active Directory domain controllers must be configured to avoid a single point of failure.
3. Every user must receive an individual username; generic shared accounts are prohibited.
4. Access must be based on job role, department and the principle of least privilege.

5. Doctors, nurses, finance staff, procurement staff, suppliers and ICT administrators must have separate access groups.
6. Administrator accounts must be separate from normal user accounts.
7. Passwords must meet the hospital password standard and multi-factor authentication must be used for administrators, remote users and suppliers.
8. Staff accounts must be disabled within 24 hours when an employee leaves, is suspended or no longer requires access.
9. Access rights must be reviewed every quarter by ICT and relevant department managers.
10. Active Directory must be backed up daily.

4.2 Linux Server SOP

Linux servers may host databases, internal web services, monitoring systems and selected hospital applications.

- Only supported long-term service releases may be used.
- Root access must be restricted to authorised ICT staff.
- Secure Shell access must use named accounts, encrypted keys and multi-factor authentication where available.
- Unused services and ports must be disabled.
- Security updates must be applied according to the patch-management schedule.
- Linux server configurations must be documented and backed up.
- Logs must be forwarded to central monitoring where possible.

4.3 Virtualisation SOP

1. Each virtual server must have a documented owner, purpose, operating system, allocated resources and backup schedule.
2. Application, database, file and monitoring systems must operate as separate virtual machines where appropriate.
3. Virtual machines must not be created without approval from the ICT Manager.
4. Snapshots must be temporary and must not replace formal backups.

5. Virtual-machine resource use must be monitored to prevent performance problems.

6. Critical virtual machines must be prioritised during recovery.

4.4 Backup and recovery SOP

The hospital must use the **3-2-1 backup principle**:

- Keep at least three copies of critical data.
- Store the copies on at least two different types of storage.
- Keep at least one copy off site or in a secure approved cloud location.



Figure 4: The 3-2-1 Backup Principle. The hospital maintains three copies of critical patient and operational data, uses at least two types of storage media, and keeps one encrypted copy off site to support recovery after equipment failure, accidental deletion or ransomware.

Data category	**Backup frequency**	**Retention**	**Recovery priority**
Electronic Patient Records	Continuous replication where possible; daily backup	According to health-record retention requirements	Critical
Hospital Management System databases	Daily full backup plus frequent transaction-log backup	Minimum 30 days online plus archival copies	Critical
Finance and payroll	Daily backup	Monthly and annual archives	High

Data category	Backup frequency	Retention	Recovery priority
Supply chain and inventory data	Daily backup	Minimum 30 days	High
Shared files and scanned patient records	Daily incremental; weekly full backup	Minimum 90 days	High
Server configurations	After every approved change	Current plus previous approved version	Critical
User computers	Approved folders backed up daily	As required	Medium

Backup procedure

1. Backup jobs must run automatically according to the approved schedule.
2. Backup data must be encrypted during transfer and storage.
3. The backup server and NAS must use controlled access and RAID protection.
4. One backup copy must be immutable or protected from deletion by ordinary administrator accounts.
5. ICT must review backup reports every morning.
6. Failed backups must be investigated within four working hours.
7. A sample restoration test must occur monthly.
8. A full recovery test for priority systems must occur twice a year.
9. Backup media or off-site backups must be stored securely.
10. No patient data may be copied to personal USB drives, private email accounts or unapproved cloud storage.

NIST guidance emphasises maintaining current software inventories, applying updates and using structured patch-management processes to reduce cyber risk.

4.5 System monitoring SOP

The monitoring platform must track:

- Server CPU, memory, disk space and storage health.
- RAID status and failed disks.
- Firewall, router, switch and Wi-Fi availability.
- Fibre-link status and internet connectivity.
- UPS power status and battery health.
- Backup-job success or failure.
- Unusual login attempts and account lockouts.
- Antivirus/EDR alerts and potential ransomware activity.
- Temperature and humidity in the server room.

Critical alerts must be sent to all three ICT support staff. The Lead ICT Technician must define after-hours escalation procedures and external vendor contact details.

Part C: Application Software Requirements SOP

5. Approved Application Software

Application	**Main purpose**	**Key users**	**Priority**
Hospital Management System	Patient registration, admission, discharge, appointments and administration	Clinical and administrative staff	Critical
Electronic Patient Record system	Patient history, diagnoses, treatment plans, prescriptions and test results	Doctors, nurses, pharmacy and authorised clinical staff	Critical
Financial Management System	Budgeting, billing, payroll, payments and reporting	Finance, HR and management	High

Application	Main purpose	Key users	Priority
Supply Chain Management System	Stock tracking, supplier orders, procurement and delivery management	Procurement, stores and supply-chain staff	High
Document Management System	Digitisation, indexing and controlled retrieval of patient and administrative files	Records staff and authorised users	Critical
Microsoft Office Suite	Documents, spreadsheets, presentations and email	Administrative and approved clinical users	Medium
Email and collaboration tools	Internal communication, calendars and approved document sharing	All approved users	High
Intralink supplier service	Communication with approved suppliers and supply-chain partners	Procurement and authorised suppliers	High
Help-desk software	Logs incidents, requests, assets and troubleshooting records	ICT staff and all hospital users	Medium

5.1 Hospital Management System and EPR SOP

1. The Hospital Management System and Electronic Patient Record system must be hosted on the application and database servers.
2. Access must be role based. Doctors, nurses, pharmacists, records staff and administrators must see only the functions and records required for their work.
3. All access to patient records must be logged.
4. Patient records may not be downloaded, emailed or copied to personal devices without formal approval.
5. The system must use automatic session timeout after inactivity.
6. Changes to diagnosis, medication, patient identity or treatment information must be auditable.

7. The system must support backup, restoration and export of data in an interoperable format.
8. Paper records created during downtime must be captured into the EPR system after restoration and checked by the responsible department.
9. The hospital must train clinical staff on correct data capture to reduce inaccurate or missing patient files.
10. The system vendor must provide service-level agreements, support contacts, security-update commitments and documented recovery procedures.

5.2 Finance and supply-chain software SOP

- Finance software must restrict payroll, payment approval and financial reporting to authorised staff.
- The system must maintain an audit trail for invoices, payments, approvals and changes.
- Supply-chain software must track stock levels, supplier orders, deliveries and expiry dates for medical supplies.
- Supplier access must be through secure VPN and multi-factor authentication.
- Suppliers may access only approved Intralink functions and must not have access to patient records, finance data or administrative network shares.
- Urgent clinical stock shortages must trigger escalation to the procurement and clinical management teams.
- Financial and supply-chain records must be backed up daily.

5.3 Microsoft Office, email and collaboration SOP

- Only licensed and approved versions of Microsoft Office and email tools may be installed.
- Microsoft Word must be used for official reports, policies and correspondence.
- Microsoft Excel may be used for controlled reporting and analysis, but not as the only storage location for critical patient records.
- Email attachments containing patient or financial information must be encrypted or shared through an approved secure system.
- Staff must not send patient files to private email accounts.

- Staff must not upload patient, employee, financial or supplier information into public AI tools or unapproved cloud applications.
- Phishing emails must be reported immediately to ICT.

Part D: Example Business Continuity Plan

6. Continuity Plan Purpose

The continuity plan ensures that the hospital can continue essential patient care when the network, fibre link, servers, systems or power supply fail. It includes technical recovery and manual processes.

NIST contingency-planning guidance includes business-impact analysis, preventative controls, recovery strategies, plan testing, training and ongoing plan maintenance.

6.1 Critical-service recovery targets

Service	**Maximum acceptable downtime**	**Recovery method**
Network core, firewall and fibre connectivity	1 hour	Use redundant firewall and activate 4G/5G backup connection
Patient registration and EPR	1 hour	Restore application/database service or use paper registration and clinical forms
Pharmacy and medication information	2 hours	Restore systems or use controlled manual prescription and stock registers
Laboratory and clinical results	2 hours	Use manual result registers until system restoration
Supply chain and procurement	8 hours	Manual purchase orders and telephone contact with approved suppliers
Finance and payroll	24 hours	Restore from backup or apply approved manual finance procedures

Service	Maximum acceptable downtime	Recovery method
Email, printing and guest Wi-Fi	24–48 hours	Restore after patient-care systems are stable

7. Example Continuity Scenario: Fibre Link Failure

Situation

The 1 km Openserve fibre link between Sekhukhune District Municipality and New Jane Furse Memorial Hospital is cut due to construction work, vandalism or equipment failure. The hospital loses its normal internet and municipal connection.

Immediate response procedure

1. A staff member reports loss of internet, EPR access or network services to ICT.
2. ICT Support Technician 1 checks the firewall, router, fibre termination device and core switch.
3. ICT confirms whether the fault is internal or an Openserve carrier outage.
4. The technician activates the 4G/5G backup router if automatic failover does not occur.
5. The Lead ICT Technician logs a fault with Openserve and records the incident reference number.
6. The hospital CEO, clinical manager and department heads are informed if clinical systems are affected.
7. Non-essential services, including guest Wi-Fi, streaming and large downloads, are restricted to preserve backup bandwidth.
8. Clinical systems receive priority through Quality of Service settings.
9. If connectivity cannot be restored within one hour, wards begin approved paper-based patient registration and clinical documentation.
10. When Openserve restores the link, ICT tests the fibre connection, firewall routing, EPR access and monitoring alerts before returning services to normal.

11. ICT captures all incident details and performs a post-incident review within five working days.

8. Example Continuity Scenario: Ransomware Attack

Situation

A staff member opens a malicious email attachment. The endpoint protection system detects suspicious encryption activity on a ward computer and alerts ICT.

Immediate response procedure

1. The user disconnects the affected computer from the network immediately.
2. ICT isolates the affected clinical or administrative VLAN if there is a risk of spread.
3. ICT blocks the affected user account and resets the password.
4. The Lead ICT Technician alerts management and the Information Officer.
5. ICT checks other servers and devices for indicators of compromise.
6. The team preserves logs and evidence; staff must not delete suspicious files.
7. If EPR or patient systems are unavailable, the clinical manager activates paper-based continuity procedures.
8. ICT rebuilds infected devices from approved standard images rather than trusting potentially compromised systems.
9. Data is restored only from clean, verified backups.
10. Before restoring network access, ICT applies patches, updates endpoint protection and confirms that no malicious activity remains.
11. The Information Officer assesses whether patient personal information was accessed or acquired by an unauthorised person and whether POPIA notification is required.
12. The hospital conducts a post-incident review, improves controls and provides targeted staff awareness training.

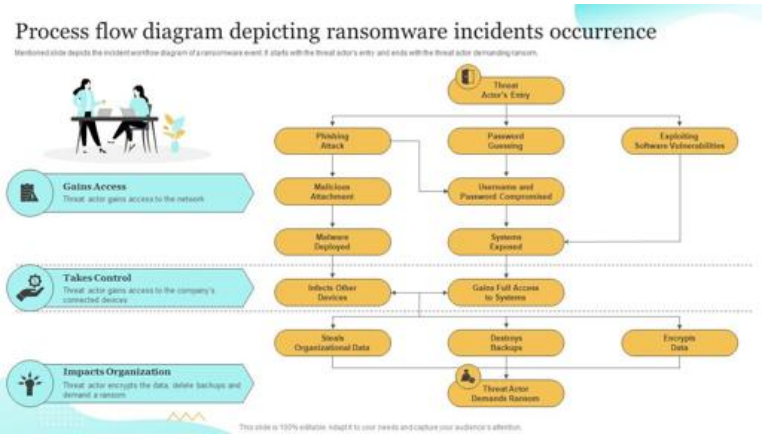


Figure 5: Example Ransomware Incident Process. Ransomware can begin through phishing or a malicious attachment, spread through systems, encrypt data and disrupt operations. The hospital's response requires rapid containment, VLAN isolation, account disabling, evidence preservation and restoration from verified backups.

9. Example Continuity Scenario: Server Failure

Situation

The primary database server fails due to disk or power-supply failure.

Recovery procedure

1. Monitoring software alerts all ICT support staff.
2. ICT checks power supply, UPS status, RAID alerts and server logs.
3. The failed server is isolated to prevent data corruption.
4. Database services fail over to the secondary database server where configured.
5. ICT confirms that the Hospital Management System and EPR are operating correctly.
6. If failover is unsuccessful, ICT restores the database from the latest verified backup.
7. The affected physical component is replaced under warranty or from spare stock.
8. ICT verifies replication and backup status after recovery.
9. The incident is recorded and reviewed to determine whether additional redundancy is required.

10. Policy and SOP Integration

The hardware, software and continuity procedures work together through the following policies:

Policy	**How it integrates with this SOP**
Access Control Policy	Active Directory, named accounts, role-based permissions, multi-factor authentication and supplier VPN controls limit unauthorised access
Network Security Policy	Firewalls, IDS/IPS, VLANs, secure Wi-Fi and network monitoring protect the fibre-connected network
Backup and Recovery Policy	Backup server, NAS, RAID storage, off-site backups and restoration tests protect patient and operational records
Acceptable Use Policy	Staff must use only approved devices, applications, email and cloud services
Asset Management Policy	ICT records all 80 desktops, 10 laptops, servers, printers, switches and software licences
Patch Management Policy	Windows, Linux, firewalls, routers, switches and applications are updated and documented regularly
Supplier Access Policy	Suppliers use restricted VPN access and may access only approved Intralink services
Incident Response Policy	Defines reporting, containment, investigation, communication and recovery after outages or cyber incidents
POPIA Privacy Policy	Ensures patient and employee personal information is collected, accessed, stored, backed up and disclosed lawfully
Business Continuity Policy	Links network and system recovery to manual procedures that protect uninterrupted patient care

This integration is important because the hospital cannot rely only on hardware. Secure servers require controlled software; software requires secure network access; patient records require backups; and all components require trained personnel, documented procedures and tested recovery plans.

Network Setup

New Jane Furse Memorial Hospital

1. Purpose of the network

The proposed network will connect the hospital's clinical, administrative, financial, supply-chain, printing, server, wireless, and approved external services. It is designed around the hardware, system/application software, and SOP requirements supplied by the group. The main objectives are to provide reliable access to the Hospital Management System and Electronic Patient Records, protect patient information, separate departments, support secure remote access, and maintain services during equipment, power, or connectivity failures.

2. Network infrastructure

The hospital will use the specified Cisco enterprise router, Cisco Catalyst 9300 managed switches, 15 PoE wireless access points, a central wireless controller, two high-availability next-generation firewalls, and the 1 km Openserve single-mode fibre connection to Sekhukhune District Municipality. (Kurose, J.F and Ross, K.W., 2021). Cat6 structured cabling, labelled patch panels, protected fibre conduit, secure racks, cooling, environmental monitoring, and UPS protection will be installed in the dedicated server room.

The eight-server environment from the group's hardware requirements will be connected to a protected server network. It includes two application servers for HMS/EPR, two database servers, a file server, backup server, RAID-enabled NAS, and an expansion server. Windows Server and Active Directory will provide central authentication and access control. (Kurose, J.F and Ross, K.W., 2021). Linux, VMware or Hyper-V, backup software, monitoring, endpoint protection, patch management, and print-management software will support the network operations described in the group SOP. (Kurose, J.F and Ross, K.W., 2021).

3. Logical network design

The municipal fibre will terminate on the Cisco router and pass through the high-availability firewall pair before reaching the core switches. The firewalls will provide IPS, VPN, SSL inspection, logging, and default-deny inbound protection. Internal traffic between departments and servers will be controlled by VLAN and firewall rules rather than being automatically trusted. (Kurose, J.F and Ross, K.W., 2021)

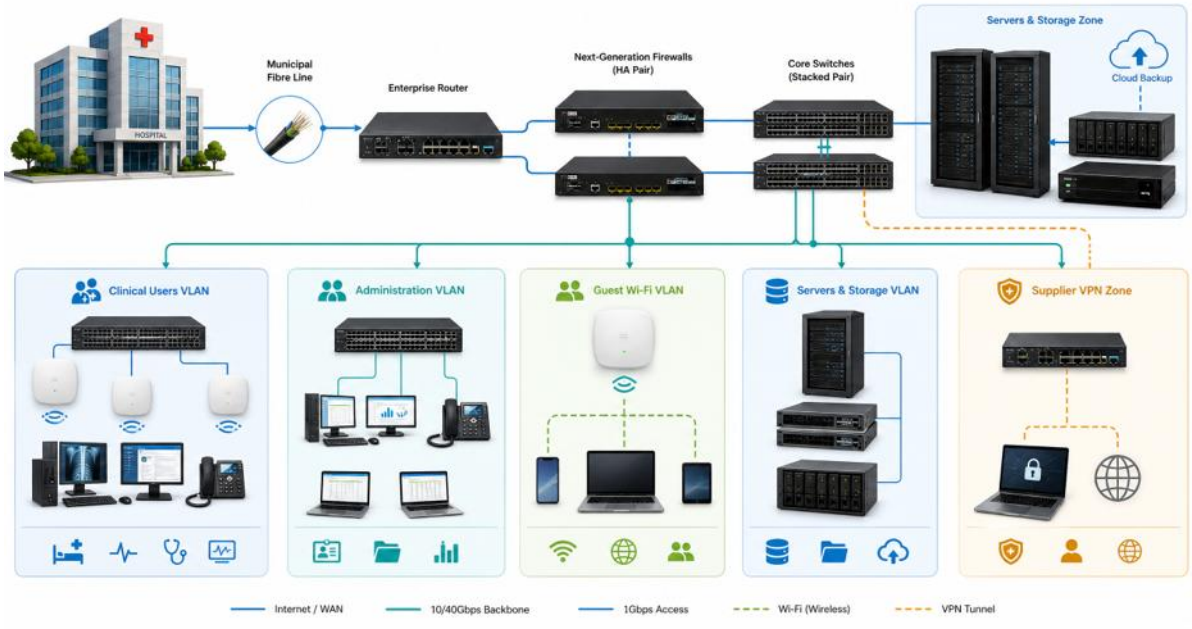


Figure 1 based logical network architecture showing the hospital, fibre edge, firewalls, core switches, severs, users, wireless, guests, suppliers, and cloud backup zones.

VLAN	Network segment	Main devices or users	Essential access control
1.	Clinical	Wards, pharmacy, laboratory, and EPR users	Wards, pharmacy, laboratory, and EPR users
2.	Administration	HR, management, records, and office users	Access to approved office, email, file, finance, and application services

3.	Finance	Finance and payroll users	Restricted to financial applications and approved file/print services
4.	Supply chain	Procurement and inventory users	Restricted to supply-chain applications and approved supplier workflows
5.	Server	Application, database, file, Active Directory, and monitoring services	No direct user access to databases; only approved application traffic
6.	Backup	Backup server and NAS	Restricted to backup and recovery operations
7.	Printers	Five network printers	Print jobs controlled through print-management software and PIN release
8.	ICT management	ICT support devices and network administration	ICT administrators only; MFA and logging required
9.	Supplier VPN	Approved external suppliers	Time-limited MFA access to named services only
10.	Guest Wi-Fi	Visitors and non-hospital devices	Internet only; no internal network access

The proposed addressing plan is 10.10.0.0/16, divided by VLAN. For example, Clinical uses 10.10.10.0/23, Administration uses 10.10.20.0/24, Server uses 10.10.50.0/24, Backup uses

10.10.60.0/24, and Guest Wi-Fi uses 10.10.100.0/23. Final addresses should be confirmed after the site survey and recorded in the network register. (Kurose, J.F and Ross, K.W., 2021).

4. Essential security controls

The following controls are required to integrate the network with the group's SOP without repeating the full policy section:

Control	Network implementation
Identity and access	Individual Active Directory accounts, role-based groups, separate administrator accounts, MFA for administrators, remote users, and suppliers, and quarterly access reviews
Firewall protection	Default-deny inbound traffic, IPS, controlled outbound access, SSL inspection where approved, VPN logging, and protected firewall administration
Network segmentation	Separate clinical, administration, finance, supply-chain, server, backup, printer, supplier, guest, and ICT management VLANs
Endpoint protection	Windows 11 devices joined to the domain, centrally patched, protected by EDR/antivirus, encrypted laptops, automatic screen locking, and no ordinary-user local administrator rights
Wireless security	Centrally managed APs, authenticated staff/clinical SSIDs, separate supplier access, guest isolation, and internet-only guest traffic
Printer security	Dedicated printer VLAN, changed default passwords, secure PIN release, controlled scan destinations, and quarterly firmware updates
Physical security	Locked racks, restricted server-room access, visitor records, cooling, smoke detection,

	environmental monitoring, and UPS protection
Monitoring	Central alerts for firewall status, fibre availability, server health, storage capacity, backup failures, endpoint security, UPS status, and temperature

5. Physical network setup

The physical installation will follow the protected path below: Sekhukhune District Municipality router → fibre termination panel → 1 km single-mode fibre in PVC or steel conduit → hospital fibre demarcation → Cisco router → HA firewalls → core switches → access switches, servers, wireless access points, computers, and printers. (Kurose, J.F and Ross, K.W., 2021).

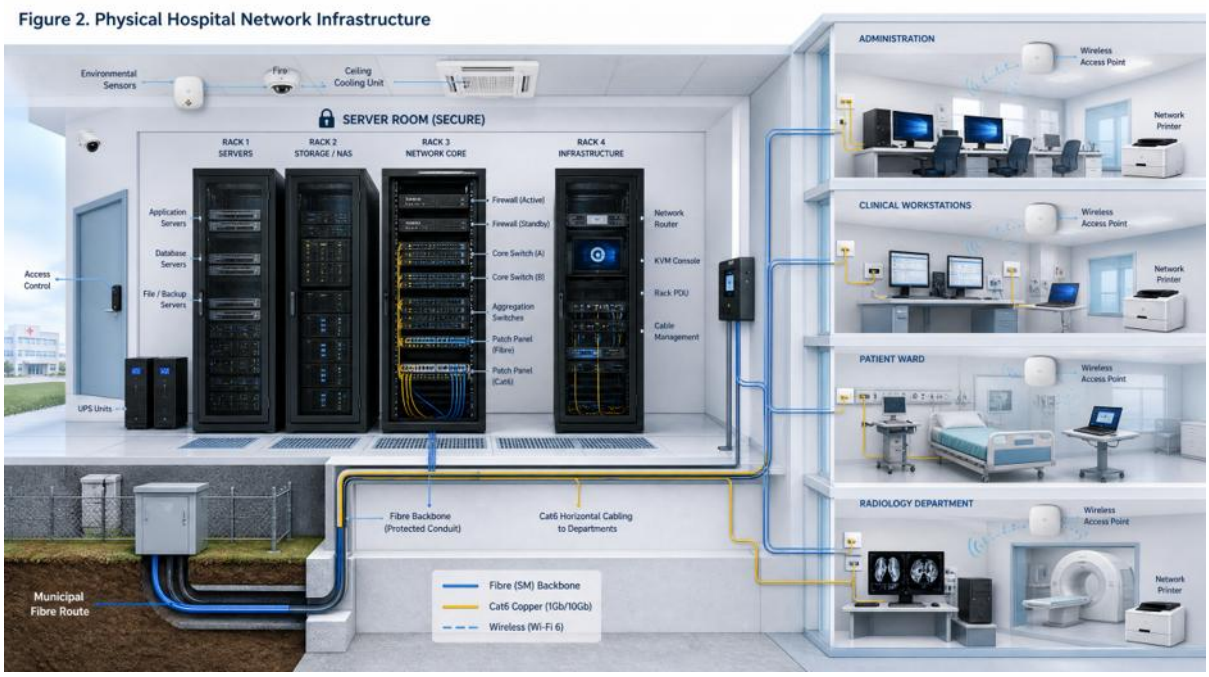


Figure 2 physical network layouts showing the secure sever room, racks, UPS, environmental controls, fibre route, Cat6 cabling, departments, wireless access points.

The server room will contain the firewall pair, router, core switching equipment, wireless controller, eight servers, NAS, patch panels, and UPS units. Access switches will connect hospital departments through certified Cat6 cabling. The 15 wireless APs will use PoE and will be placed according to a wireless site survey. (Kurose, J.F and Ross, K.W., 2021). All equipment,

ports, cables, IP addresses, serial numbers, owners, warranties, and backup status must be recorded in the ICT asset register.

6. Cloud, municipal, and supplier connections

The hospital's primary HMS/EPR data will remain on the protected internal server network. An approved secure cloud or off-site repository may be used for an encrypted backup copy or disaster-recovery workload after approval by hospital management and the Information Officer. (Kurose, J.F and Ross, K.W., 2021). Email, collaboration, software updates, and approved external services will pass through the firewall using HTTPS and identity controls.

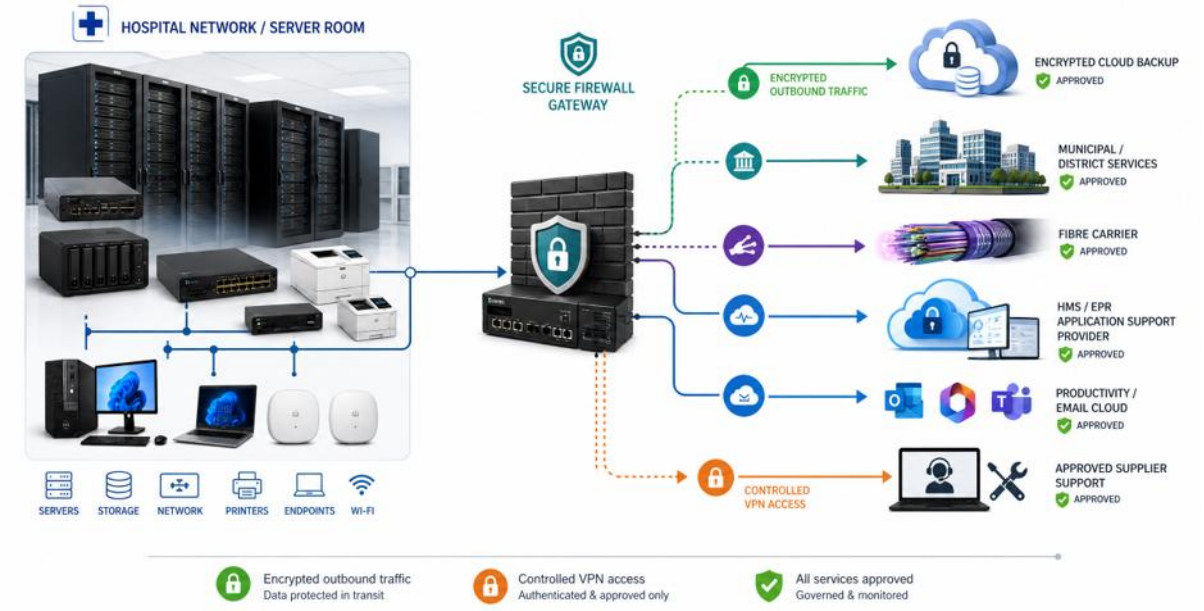


Figure 3 view of approved cloud, municipal, carrier, application-support ,productivity and supplier interactions through the secure firewall gateway.

External service	Permitted connection	Required protection
Openserve	Fibre transport and service support.	Circuit monitoring, documented demarcation, and carrier escalation procedure.
Municipality	Approved district services.	Router ACLs, firewall rules, and named technical contacts.

HMS/EPR supplier	Application support.	MFA VPN, named account, time-limited access, logging, and no general database access.
Approved cloud/off-site backup	Encrypted backup or disaster-recovery copy.	Encryption, retention, immutability, privacy approval, and restore testing.
Productivity provider	Email, collaboration, Office licensing, and updates.	MFA, HTTPS, identity management, and approved data handling.

7. Backup and continuity

The network will follow the SOP's 3-2-1 backup principle: production data on resilient server storage, a second encrypted copy on the backup server/NAS, and a third protected copy off-site or in an approved cloud repository. (Kurose, J.F and Ross, K.W., 2021). At least one copy must be immutable or protected from deletion by ordinary administrators. Backup and restoration must be tested at least twice a year.

UPS units will protect servers, firewalls, switches, and other critical equipment for the specified two-to-four-hour target runtime, subject to the final electrical load calculation. During an outage, clinical services receive priority. A fibre failure requires carrier escalation and use of the approved alternative communication or offline procedure. (Kurose, J.F and Ross, K.W., 2021). A ransomware incident requires immediate isolation of affected devices, account protection, incident reporting, verification of clean backups, and controlled restoration.

1. Network implementation and testing

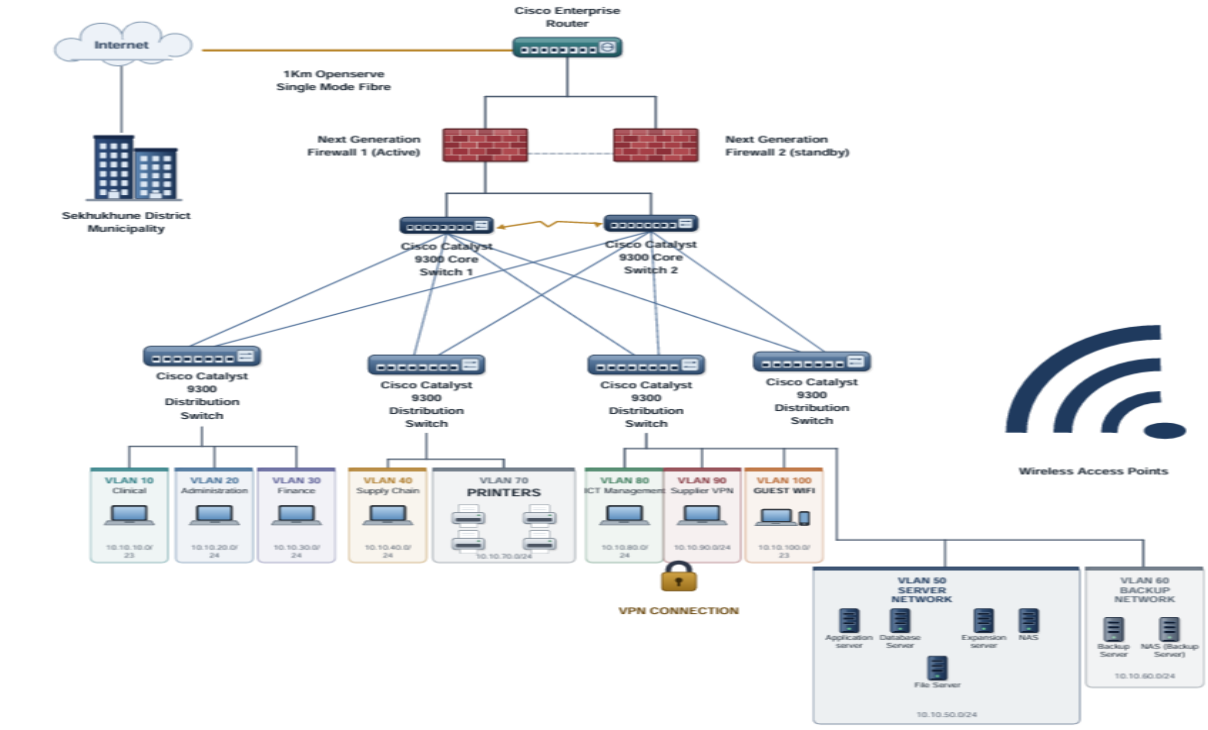
Stage	Main activity	Evidence required
1.	Complete site survey and confirm rack, cabling, fibre, power, and AP locations.	Approved design and site survey.

2.	Install server room, racks, UPS, Cat6 cabling, fibre, and patch panels,	Cable, fibre, UPS, and environmental test results.
3.	Install router, HA firewalls, core switches, VLANs, VPN, IPS, and logging.	Firewall, failover, and segmentation test results.
4.	Install servers, Active Directory, virtualisation, applications, backup, monitoring, and endpoint security.	Service, access, monitoring, and backup tests.
5.	Connect computers, laptops, printers, and wireless Aps.	Device, wireless coverage, and secure-print tests.
6.	Perform handover and user acceptance testing.	Updated diagrams, asset register, SOP, training, and sign-off.

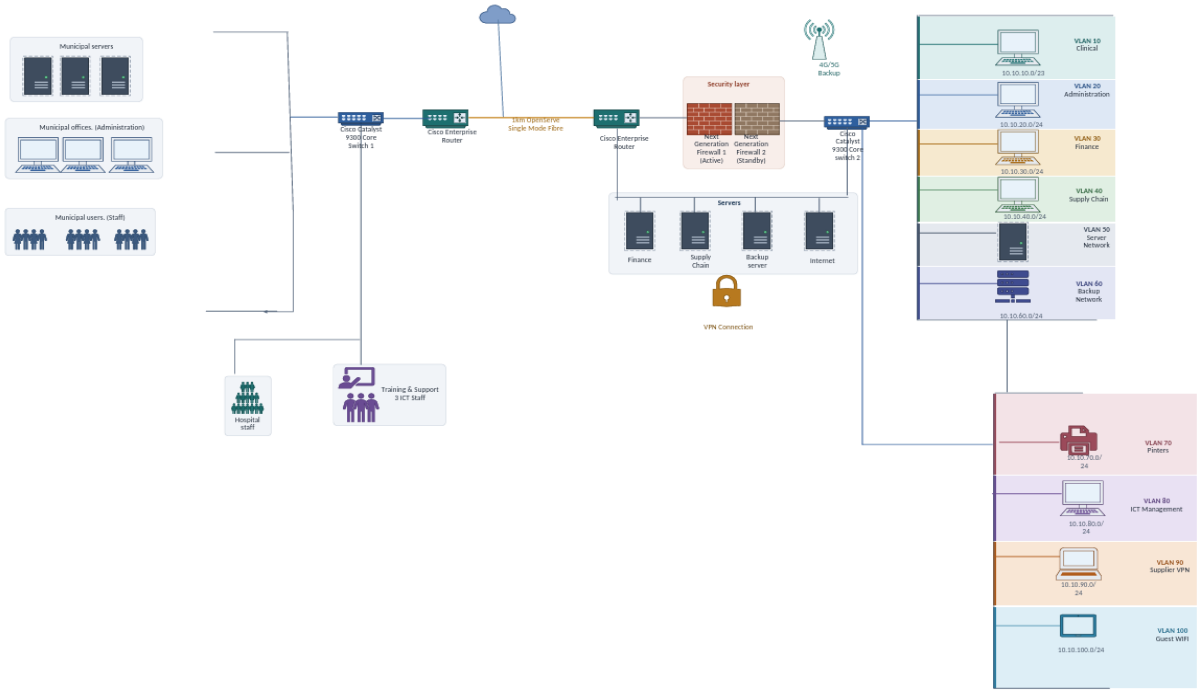
The final acceptance test must confirm that guest users cannot reach internal systems, suppliers can reach only approved services, clinical users can access EPR according to their roles, ordinary users cannot directly access databases, firewall failover works, backups can be restored, printers require PIN release, wireless coverage is sufficient, and monitoring alerts reach the ICT team. (Kurose, J.F and Ross, K.W., 2021).

This network setup includes the essential network-related requirements from the group's hardware, software, and SOP documents without repeating their full sections. It provides the required infrastructure, segmentation, security, physical installation, external connections, continuity controls, and implementation testing needed for a complete and neat network setup report.

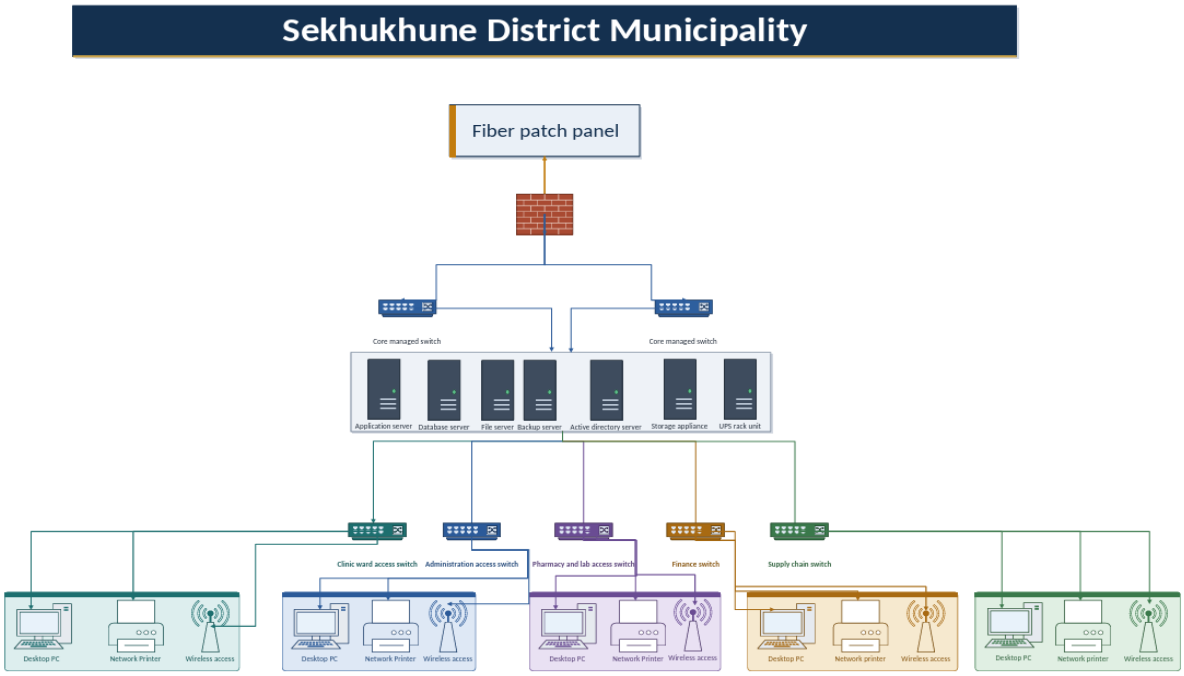
Network diagram



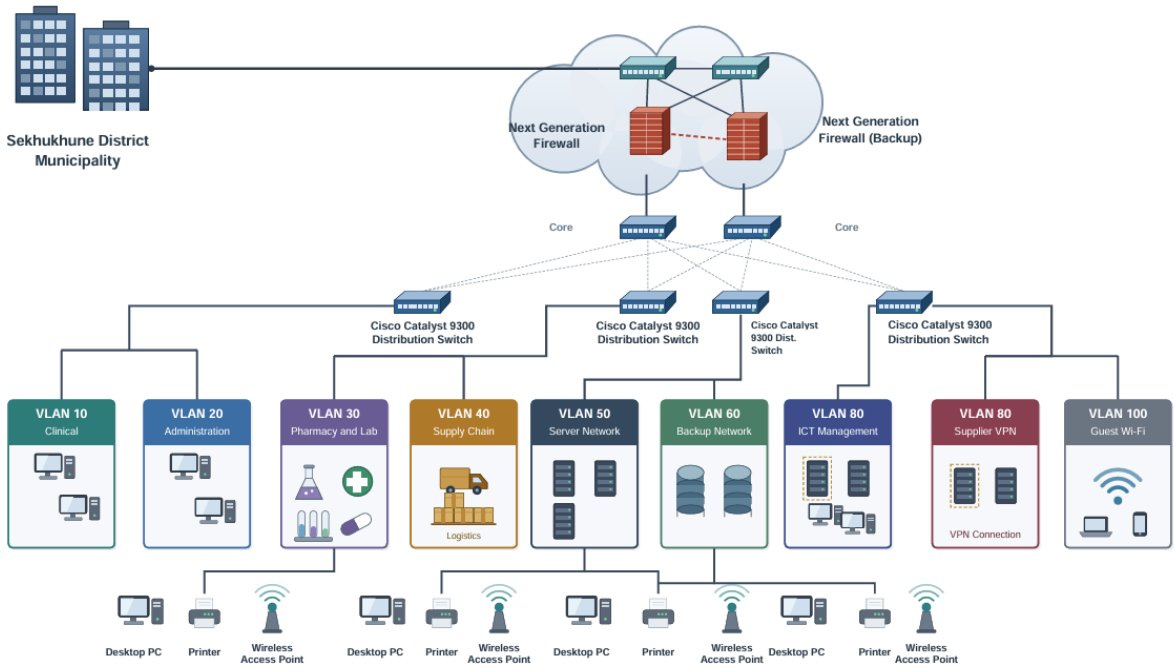
Logical diagram



Physical diagram



Cloud diagram



Outsourced services

Given the remote location of the New Jane Furse Memorial Hospital, the severe constraint of having only three internal ICT support staff, and the requirement for a self-sustaining yet world-class network, the hospital must strategically leverage outsourced services. The project team has identified several key areas where external expertise is critical to bridge the digital divide and ensure the long-term viability of the infrastructure.

The most foundational outsourced service is the Carrier and Fibre Connectivity provided by Openserve. As identified in the project requirements, the hospital requires a dedicated 1 km fibre-optic connection to splice from the Sekhukhune District Municipality network. Because laying and terminating fibre-optic cable requires specialized equipment, wayleave approvals, and deep trenching expertise, this cannot be handled internally. Outsourcing to Openserve as the primary carrier ensures the physical layer of the network is professionally installed, tested, and maintained. Furthermore, because the hospital's three ICT staff cannot be expected to repair fibre cuts or manage municipal routing, outsourcing this service guarantees carrier-grade reliability and uptime, which is essential for continuous access to the Electronic Patient Record (EPR) system.

To support the highly technical server and security environment without overburdening the internal team, Managed Cybersecurity and Firewall Services must be outsourced. The hospital currently lacks security measures and is vulnerable to ransomware and data breaches. The proposed solution includes dual next-generation firewalls, Intrusion Detection/Prevention Systems (IDS/IPS), and VPN gateways for supplier access. Managing these advanced security protocols requires 24/7 monitoring and specialized threat intelligence that three ICT staff cannot provide. By outsourcing security monitoring to a Managed Security Service Provider (MSSP), the hospital ensures that suspicious activity, such as unauthorized access attempts to patient records is detected and mitigated immediately, fulfilling POPIA compliance requirements while allowing internal staff to focus on daily operational support.

Application Support and Cloud Hosting represent another critical outsourced component. The hospital relies heavily on the Hospital Management System (HMS) and EPR software. Hosting and maintaining these applications internally would require extensive database administration skills that are currently scarce at the hospital. Therefore, the project proposes outsourcing the hosting of the HMS/EPR to a secure, approved cloud environment or relying on the software vendor for application support and updates. This is essential for maintaining data integrity and ensuring that the system is backed up off-site the 3-2-1 backup principle to prevent total data loss in the event of a local disaster or server theft.

Finally, to address the skills gap and ensure the long-term sustainability of the project, Specialized Technical Training and Maintenance Contracts will be outsourced. While the project aims to train the three internal ICT staff, they cannot be expected to master complex Cisco routing, VMware virtualization, and advanced troubleshooting overnight. The project will outsource specialized training to external vendors to upskill the staff on the specific hardware installed, as well as establish maintenance agreements for the hardware (servers, switches, UPS). This ensures that should a critical component fail, such as a core switch or a database server, expert support is available to assist the internal team in replacing the unit, thereby reducing downtime and ensuring the network remains a sustainable asset for the Sekhukhune District.

Other IT Support

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1. Project Title

Improving the hospital's IT infrastructure and network system

The project aims to improve the ICT infrastructure at New Jane Furse Memorial Hospital by upgrading computer equipment, setting up a secure and reliable network, improving data storage and providing the systems needed to support the hospital's daily operations.

2. Project Purpose

The purpose of this project is to improve the ICT infrastructure at New Jane Furse Memorial Hospital in the Sekhukhune District of Limpopo. The hospital currently has outdated computers, limited printing equipment, poor patient record storage and network infrastructure that needs to improve. These challenges are making it difficult for staff to access information, manage patient records and do daily administrative and clinical tasks well.

The project will solve these problems by providing updated computers and printers, reliable network connectivity, secure data storage, suitable hospital information systems and improved cybersecurity measures. It will also include a dedicated server room, backup facilities and a fibre connection to the Sekhukhune District Municipality.

The project is needed to support more reliable access to hospital information, protect sensitive patient data, improve communication and help staff perform their duties more effectively. The proposed improvements will provide a stronger ICT foundation for the hospital's current operations and future needs.



Figure 1: Purpose and justification of the hospital ICT project.

3. Project Description

The project involves upgrading the ICT infrastructure at New Jane Furse Memorial Hospital in the Sekhukhune District of Limpopo. The aim is to provide reliable computer equipment, secure network connectivity, improved data storage and suitable information systems to support the hospital's daily operations.

The project will involve assessing the hospital's current ICT infrastructure, preparing the server room, installing network equipment and structured cabling and establishing a fibre connection to the Sekhukhune District Municipality. It will also include installing and configuring servers, computers, laptops, printers, hospital information systems and cybersecurity software.

The project will include setting up network access for different hospital departments, setting up data backup and recovery measures, testing the new infrastructure, training users and handing over the completed system to the hospital's ICT team.

The major project work items are:

1. **ICT assessment and planning:** Review the existing equipment, network, power supply and hospital ICT needs.
2. **Server room preparation:** Prepare the server room and install racks, cooling equipment, environmental monitoring and an uninterruptible power supply (UPS).
3. **Network installation:** Install structured cabling, network switches, firewalls, wireless access points and the fibre connection.
4. **Server and software installation:** Set up servers, data storage, hospital information systems, user accounts, security software and backup facilities.
5. **End-user equipment installation:** Install and configure desktop computers, laptops and network printers.
6. **Testing and security checks:** Test network connectivity, system access, data backups and security controls.
7. **Training and handover:** Train hospital staff, provide updated network diagrams and operating procedures and obtain final approval from the relevant stakeholders.

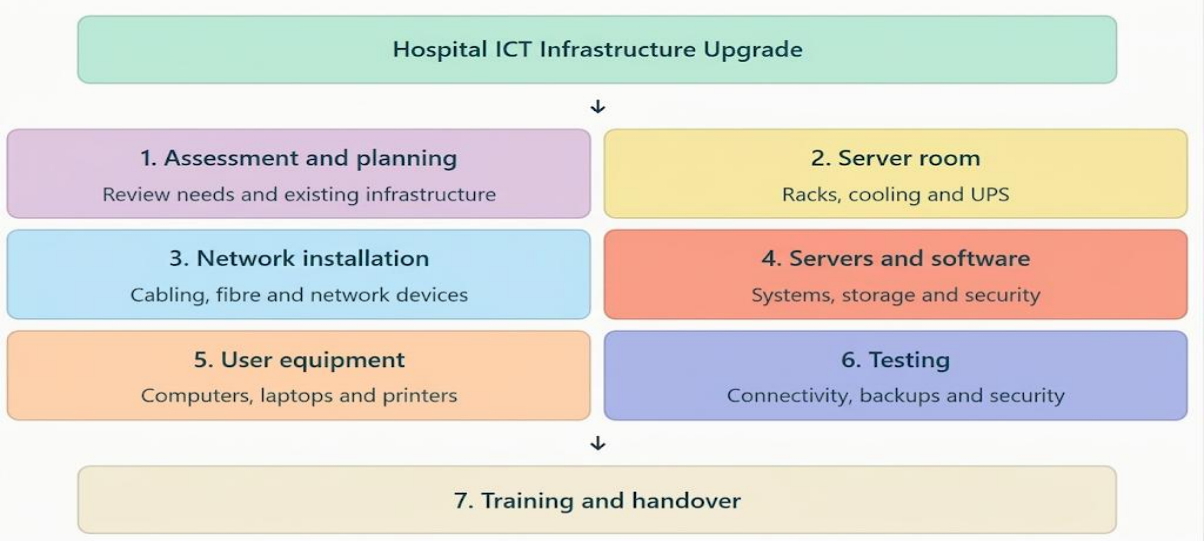


Figure 2: Preliminary Work Breakdown Structure for the hospital ICT project

4. Project Objective

The main objective of this project is to design and set up a secure, reliable, and sustainable network infrastructure connecting New Jane Furse Memorial Hospital to the Sekhukhune District Municipality network within 12 months.

The project aims to install a dedicated 1 km fibre-optic connection, upgrade the hospital's network equipment, improve data storage, strengthen cybersecurity, provide secure wireless connectivity, improve communication with suppliers and train the hospital's three ICT support staff members to maintain the new infrastructure.

The project is expected to be completed within the estimated budget of R5 335 000. Successful completion will be measured by reliable network connectivity, successful testing of the installed infrastructure, improved protection of hospital information, completion of staff training and formal handover of the system.

Project target	Details
Completion time	12 months
Estimated budget	R5 335 000
Main network connection	1 km fibre-optic link
Staff training	3 hospital ICT staff members
Final outcome	Tested, secure network infrastructure formally handed over

Figure 3: Key project objectives and targets.

5. Success Criteria or Expected Benefits

The success of the ICT Network, Cybersecurity and Fibre Connectivity Project will be measured by whether the new infrastructure meets the hospital's needs and improves its daily operations. The project is expected to provide reliable network connectivity, improve the storage and protection of hospital information and support communication between hospital departments and suppliers.

The project will be considered successful when the 1 km fibre-optic connection between the hospital and the municipal network has been installed and tested, the new network equipment and servers are working correctly, and the required cybersecurity measures and data backups are in place. The hospital's three ICT support staff members must also complete training so they can maintain and support the new infrastructure.

The project must be completed within the approved 12-month period and the estimated budget of R5 335 000. Final acceptance will depend on successful testing, confirmation that the agreed project requirements have been metand formal handover to hospital management.

Success criterion	How success will be measured
Fibre connectivity	The 1 km fibre-optic link is installed and successfully tested.
Network equipment	Installed network equipment and approved end-user devices are operational.
Data protection	Secure data storage and backup and recovery measures are tested.
Network reliability	Network connectivity is tested across the required hospital departments.
Staff training	All 3 hospital ICT support staff members complete the planned training.
Budget	Project expenditure stays within the approved R5 335 000 budget.
Completion time	Project is completed within the planned 12 months.
Handover	Testing is completed and the infrastructure is formally handed over to the hospital.

6. Funding

The estimated funding required for the New Jane Furse Memorial Hospital ICT Infrastructure and Network Project is R5 335 000. This amount includes the installation of the fibre-optic connection, network equipment, computers and printers, servers and storage, cybersecurity measures, structured cabling, staff training, testing and documentation, as well as the contingency allowance.

The project sponsor, Sekhukhune District Municipality, is responsible for authorising the project funding. The funds should be managed according to the approved project budget and used for the planned project activities. Payments and spending should be monitored throughout the project to ensure that costs remain within the approved amount.

If funding is released in stages, further funding may be linked to satisfactory progress and the completion of agreed project milestones. The exact funding release schedule must be confirmed by the project sponsor.

Budget item	Estimated amount
Fibre-optic connection	R1 200 000
Network equipment	R850 000
Computers and printers	R900 000
Servers and storage	R750 000
Cybersecurity	R500 000
Structured cabling and installation	R400 000
Staff training	R150 000
Testing and documentation	R100 000
Contingency allowance	R485 000
Total estimated funding	**R5 335 000**

Table : Estimated Project Funding Breakdown

7. Major Deliverables

The major deliverables for the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project will include 80 desktop computers, 10 laptops, 15 network printers, an improved hospital network and secure access to hospital systems. The project will also provide improved data storage, regular backups and staff training to help hospital employees use the new systems. These deliverables will help the hospital manage patient information more securely, reduce missing records and improve communication and daily operation

8. Acceptance Criteria

The acceptance criteria are the requirements that each project deliverable must meet before the project sponsor can approve it. These criteria will be used to check that the hospital's new ICT equipment, network and security systems work correctly and meet the project's requirements.

For the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project, the acceptance criteria will include the following:

- **Desktop computers:** All 80 desktop computers must be installed, tested and working correctly.
- **Laptops:** All 10 laptops must be installed with the required software and tested before being handed over to hospital staff.
- **Network printers:** All 15 network printers must print successfully and be accessible to authorised hospital staff.
- **Hospital network:** The upgraded network must provide reliable connectivity to the required hospital departments, with network access tested before acceptance.
- **Security and data backups:** User access controls and multi-factor authentication must be tested. Data backups must be completed successfully, and a test restoration must confirm that backed-up data can be recovered.
- **Staff training:** Staff members assigned to training must complete the planned training, and their ability to use the new systems must be checked.
- **Final handover:** The project sponsor must review the test results and confirm that the agreed requirements have been met before formally accepting the deliverables.

9. Milestone Schedule

The milestone schedule lists the important events in the project and the target time for completing each one. It will help the project team monitor progress and ensure that the hospital's ICT infrastructure upgrade is completed within the planned 12-month period.

The following milestones are proposed for the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project.

Milestone	Target completion
Project planning and approval	Month 1
Assessment of the existing ICT infrastructure	Month 2
Network design and equipment specifications	Month 3
Procurement of computers, laptops, printers and network equipment	Month 4
Network cabling and connectivity improvements	Month 6
Installation of desktop computers, laptops and network printers	Month 8
Network configuration, VLANs and security implementation	Month 9
Data backup, system testing and troubleshooting	Month 10
Staff training and user testing	Month 11
Final inspection, project acceptance and handover	Month 12

10. Key Assumptions

The key assumptions are the conditions that are expected to be true for the project to proceed as planned. The New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project is based on the following assumptions:

1. **Funding availability:** The approved project budget of R5,335,000 will be available to cover the planned ICT equipment, network improvements, security measures and implementation costs.
2. **Equipment availability:** The required desktop computers, laptops, network printers and network equipment will be available from suppliers within the planned procurement period.

3. **Network connectivity:** Openserve fibre connectivity from the municipality will be available and suitable for the hospital's network requirements.
4. **Staff availability:** The hospital's ICT team and relevant hospital staff will be available to assist with installation, testing and training.
5. **Management support:** Hospital management will approve the necessary project activities and provide access to the departments where the ICT upgrades will take place.
6. **Minimal disruption:** The installation and upgrade work can be completed while allowing the hospital to continue providing essential healthcare services.
7. **Security and system requirements:** The hospital will be able to implement the planned network security measures, including VLANs and multi-factor authentication, using suitable equipment and software.
8. **Project timeframe:** The required resources, approvals and equipment will be available to support completion of the project within the planned 12-month period.

11. Project Constraints

Project constraints are the limitations or restrictions that may affect the planning and implementation of a project. The New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project will have the following constraints:

1. **Budget limitation:** The project must be completed within the approved budget of R5,335,000.
2. **Time limitation:** The project must be completed within the planned 12-month period.
3. **Hospital operations:** The installation and upgrading of ICT equipment must be planned to avoid disrupting essential healthcare services and access to patient information.
4. **Limited ICT staff:** The hospital has only three ICT staff members to support its existing systems and assist with the upgrade. Their availability may affect the pace of implementation.
5. **Existing infrastructure:** The project team must work with the hospital's existing buildings, network layout and infrastructure when installing new equipment and improving connectivity.

6. **Equipment procurement:** The project schedule depends on suppliers delivering the required computers, laptops, printers and network equipment on time.
7. **Data security and privacy:** All work involving patient information must protect confidential records and comply with applicable data protection requirements.
8. **Staff availability:** Hospital employees must continue performing their normal duties while attending training and participating in system testing.

12. Major Risks

Major risks are events or problems that could seriously affect the completion of a project. The New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project may face the following major risks:

1. **Budget overruns:** The cost of ICT equipment, network installation or security systems may be higher than expected, causing the project to exceed its approved budget of R5,335,000.
2. **Supplier delays:** Delays in delivering computers, laptops, printers and network equipment could affect the project schedule and prevent the team from completing the upgrades within 12 months.
3. **Network installation problems:** Difficulties with fibre connectivity, network cabling or equipment configuration could delay implementation and increase costs.
4. **Data loss:** Errors during system upgrades or data migration could result in the loss or corruption of important patient records.
5. **Cybersecurity threats:** Cyberattacks or unauthorised access during implementation could expose confidential patient information and affect hospital systems.
6. **Disruption to hospital services:** Installing new equipment or upgrading the network could interrupt access to patient information and affect the daily work of hospital staff.
7. **Limited ICT capacity:** The hospital's three ICT staff members may not have enough time or capacity to manage the existing systems while supporting the upgrade, which could cause delays.

8. **Staff resistance or limited training:** Some employees may struggle to adapt to the new systems if they do not receive sufficient training, reducing the benefits of the upgrade.

These risks will need to be monitored throughout the project. The project manager should plan preventive measures, arrange suitable technical support and address problems early to reduce their effect on the project's budget, schedule and objectives.

Table 12.1: Project Risk Assessment

Identified risk	Likelihood	Impact	Risk level	Recommended response
Budget overruns	Medium	High	High	Monitor spending and keep costs within the approved R5,335,000 budget.
Supplier delays	Medium	High	High	Order equipment early and follow up with suppliers regularly.
Network installation problems	Medium	High	High	Test the network equipment and connectivity during installation.
Data loss	Low	High	High	Back up patient records and test data restoration before making major changes.
Cybersecurity threats	Medium	High	High	Implement access controls, multi-factor authentication and network security measures.
Disruption to hospital services	Medium	High	High	Schedule installation work to minimise interruptions to healthcare services.
Limited ICT staff capacity	High	Medium	High	Plan staff workloads and arrange additional technical support if needed.
Staff resistance or limited training	Medium	Medium	Medium	Provide training and support to help staff use the new systems correctly.

13. Approval Requirements

Approval requirements define who is authorised to make decisions and approve activities during the project. For the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project, the following approval requirements will apply:

1. **Project approval:** Hospital management or the appointed project sponsor must approve the project charter before the project begins.

2. **Budget approval:** The project sponsor must approve the project budget of R5,335,000. Any proposed spending that exceeds the approved budget must be submitted for further approval.
3. **Purchasing approval:** The project manager may coordinate the purchasing of ICT equipment within the approved budget and delegated authority. Purchases outside these limits must be approved by the authorised hospital management representative.
4. **Changes to the project:** Major changes to the project scope, budget or 12-month schedule must be reviewed and approved by the project sponsor before implementation.
5. **Phase completion approval:** The project sponsor must review the results of important project stages, including equipment installation, network implementation and system testing, before approving the next stage where required.
6. **Security and data protection approval:** The relevant hospital management and authorised ICT personnel must approve the required security arrangements before the new systems are used to handle patient information.
7. **Final project acceptance:** The project sponsor must review the completed deliverables and test results and formally approve the project handover before the project is considered complete.

These approval requirements will help ensure that project decisions are properly authorised, expenditure is controlled and the completed ICT infrastructure meets the hospital's requirements.

14. Project Manager

The project manager is the person appointed to plan, organise, lead and monitor the project to ensure that it meets its objectives within the approved budget and timeframe.

For the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project, the project manager will be responsible for coordinating the project team, managing the 12-month schedule, monitoring the approved budget of R5,335,000 and ensuring that the required ICT equipment and network improvements are delivered.

The project manager will also coordinate with hospital management, the ICT team and equipment suppliers, monitor project risks, report progress to the project sponsor and ensure that the required approvals are obtained. The project manager will oversee the installation and testing of the new systems and coordinate staff training and the final handover.

The appointed project manager will work with the project sponsor and relevant hospital staff to develop the project plan and ensure that the project is completed according to the agreed requirements.

15. Reporting Requirements

Reporting requirements specify how often project progress must be reported, what information must be included and who must receive the reports.

For the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project, the project manager will prepare monthly written progress reports and submit them to the project sponsor and hospital management. These reports will provide updates on the project's progress against the planned 12-month schedule and the approved budget of R5,335,000.

The reports will include the following information:

1. **Project progress:** Completed activities and milestones compared with the project schedule.
2. **Budget status:** Expenditure to date and any expected cost increases.
3. **Major risks and issues:** Problems that could affect the project timeline, budget, security or successful completion.
4. **Equipment and procurement:** Updates on equipment orders, deliveries and installation.
5. **Network and security implementation:** Progress with network improvements, system testing, data backups and security measures.
6. **Staff training:** Updates on training activities and staff readiness to use the new systems.
7. **Planned activities:** Work scheduled for the next reporting period.

8. **Decisions required:** Any issues that require the project sponsor's approval or intervention.

In addition to the monthly reports, the project manager will arrange quarterly review meetings with the project sponsor and relevant hospital representatives to discuss progress, address major concerns and approve any necessary corrective actions. Urgent risks or issues affecting patient information or essential hospital services must be reported to the appropriate hospital authorities immediately.

16. Sponsor Designee

The sponsor designee is the person appointed by the project sponsor to represent them and oversee certain project activities. The project manager communicates with the sponsor designee, reports on project progress and remains accountable to them for the responsibilities assigned.

For the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project, the hospital management representative appointed by the project sponsor will act as the sponsor designee. This person will receive the monthly progress reports, review the project schedule and budget, monitor major risks and assist with decisions that require management involvement.

The sponsor designee will also communicate the sponsor's requirements to the project manager and help ensure that the project remains aligned with the hospital's needs. If authorised by the project sponsor, the designee may approve completed deliverables and confirm that they meet the agreed acceptance criteria.

The sponsor designee will work closely with the project manager and hospital ICT team to support the successful completion and handover of the upgraded ICT infrastructure.

17. Approval Signature and Date

The approval signature and date confirm that the project sponsor has officially authorised the New Jane Furse Memorial Hospital ICT Infrastructure Upgrade Project. Once the Project Charter has been reviewed and approved, the project can proceed according to the agreed objectives, budget and schedule.

The approval date will serve as the official project start reference for tracking the target dates of the key milestones throughout the planned 12-month implementation period.

Project Sponsor: _Tlotlang Ramatlotlo_____

Position: _Hospital Management Representative_____

Signature: 

Date: _____2026/10/21_____

Reference list

Gido, J., Clements, J. and Baker, R. (2018) Successful Project Management. 7th ed. Boston, MA: Cengage Learning.